AI Playbook and Toolkit for Public Health Departments

Managing AI Projects in Public Health Programs

PGM 100

This module introduces program managers to the practical responsibilities involved in managing AI projects in public health settings. It focuses on problem definition, governance intake, stakeholder coordination, approved-tool awareness, workflow fit, risk review, implementation planning, and monitoring. It is written for national use, so it does not assume a specific agency tool, procurement process, or governance structure.

Level	introductory/applied
Primary track	Program Management Track
Appears in tracks	program-management

Learning Objectives

- Describe the program manager role in responsible AI project planning and implementation.
- Distinguish between tool-first ideas and workflow-based public health use cases.
- Identify the governance, data, workforce, vendor, equity, privacy, and monitoring questions that must be addressed before implementation.
- Create a project framing brief for a proposed AI-supported public health workflow.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level training module is intended for public health education, planning, and workforce development. It does not replace agency policy, legal review, privacy review, security review, procurement review, communications clearance, civil rights review, Tribal consultation, or governance approval. Learners should confirm applicable requirements, approved tools, data rules, and review pathways within their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Many AI projects fail or create risk because they begin as technology demonstrations rather than managed public health projects. Program managers are often the people who must turn an idea into a defined problem, a feasible workflow, an accountable team, and an implementation plan. They do not need to be data scientists, but they need enough AI literacy to ask the right questions and route the project to the right reviewers. Responsible AI project management begins with governance intake and continues through monitoring and lifecycle decisions.

Public health AI projects often involve multiple teams, including programs, epidemiology, IT, legal, privacy, security, procurement, communications, equity, community engagement, and leadership. Without coordination, projects may move forward with unclear ownership, unapproved tools, poor data readiness, unrealistic timelines, or no plan for human review. Program managers help prevent these failures by making assumptions, decisions, dependencies, and review needs visible. They also ensure that AI work remains connected to public health value rather than novelty.

Public Health Example

A chronic disease program wants to use AI to identify communities that may benefit from outreach about hypertension screening. The program manager first documents the public health purpose, data sources, affected populations, intended users, proposed workflow, human review points, and governance questions. Instead of asking a vendor to build a model immediately, the manager routes the idea through intake and triage to determine data readiness, privacy review, equity concerns, and whether a simpler analytic approach might be sufficient. This helps the team decide whether the project should proceed, change scope, or wait for additional preparation.

Technical, Programmatic, and Operational Deep Dive

A program manager should begin by clarifying the problem and the decision that AI would support. The project framing brief should explain who experiences the problem, why current workflows are insufficient, and what improvement would matter. It should also identify whether AI is necessary or whether a policy, staffing, workflow, data quality, or communication change would solve the problem more directly. This protects the agency from investing in AI when the underlying issue is not technical.

The next step is to map stakeholders and review needs. Program managers should identify business owners, technical owners, data stewards, intended users, affected groups, and reviewers for privacy, security, legal, equity, procurement, communications, and governance. They should also determine whether the project involves public-facing outputs, decision support, sensitive data, automation, external vendors, or community impact. These factors determine the level of review required before implementation.

AI project management also requires planning beyond the pilot. The project plan should address training, change management, user acceptance testing, validation, monitoring, incident reporting, funding, sustainment, and retirement criteria. Program managers should document what evidence will be used to decide whether the project should scale, revise, pause, or stop. This makes AI implementation accountable rather than experimental by default.

Risks, Failure Modes, and Guardrails

A major failure mode is allowing a project to become vendor-led before the public health problem is defined. Vendors may demonstrate impressive capabilities, but the agency remains responsible for purpose, data use, equity, privacy, and public impact. Guardrails include use case intake, governance triage, internal requirements, and clear decision rights before vendor engagement. Program managers should ensure that vendor conversations do not replace public health problem definition.

Another risk is underestimating workflow and workforce impacts. AI tools may add review steps, change roles, alter documentation, or create new monitoring responsibilities. If staff are not prepared, the tool may be ignored, misused, or overtrusted. Guardrails include workflow mapping, training plans, supervisor readiness, and feedback loops before and during implementation.

A third risk is treating project launch as success. An AI-supported workflow can be launched and still fail to improve outcomes, reduce burden, or maintain equity. Program managers should require monitoring metrics, user feedback, and governance review after go-live. The project should have explicit criteria for continuation, revision, pause, or retirement.

Application to AI-Supported Workflows

This module applies to any AI-supported workflow being proposed, piloted, procured, implemented, or evaluated by a public health program. The program manager should use intake and project framing to define scope, owners, review needs, and evidence requirements. The project should not proceed to procurement or technical build until the team understands data, risk, workflow, human review, and monitoring expectations. Approved-tool awareness should be part of the project plan because teams must know which tools are authorized for which data and tasks.

Reflection Questions

What AI ideas in your program are actually workflow problems that need better definition?

Who needs to review an AI project before a pilot or vendor discussion begins?

What evidence would show that an AI project is worth scaling or continuing?

Practical Exercise

Create a project framing brief for a proposed AI-supported public health program workflow. Include the public health problem, proposed AI support, users, data, affected populations, owners, review needs, approved-tool considerations, risks, expected benefits, monitoring metrics, and next decision point.

Example: For an AI-supported referral follow-up workflow, the brief might explain that staff need help prioritizing follow-up calls, identify referral data as sensitive, require human review before outreach decisions, include equity review for missed populations, and define success as improved timely contact without reducing service access for lower-data groups.

Expected Artifact or Evidence

AI project framing brief.

Stakeholder and reviewer list.

Initial monitoring and decision criteria.

Knowledge Check

- 1. What should come before vendor selection in an AI project?
- 2. Why do program managers need AI literacy?
- 3. What is a tool-first project?
- 4. What should be planned beyond the pilot?
- 5. What is one appropriate program manager guardrail?

References and Resources

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